



PROPOSAL A2 — GOVERNMENT ROAD DEFECT REPORTING APP

AI-Enabled Citizen Infrastructure Monitoring & Road Maintenance Governance Platform

Prepared for: Project TN, Government of Tamil Nadu

Submitted by: VECTRO DIGITAL SERVICES PRIVATE LIMITED + EXtream.AI

Proposal Version: 1.0

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CONFIDENTIALITY NOTICE

This document contains strategic governance architecture, citizen reporting workflows, AI-enabled analytics concepts, operational models, and deployment recommendations intended solely for Project TN leadership, Government of Tamil Nadu stakeholders, and authorized evaluation committees.

1. EXECUTIVE SUMMARY

The Government Road Defect Reporting App is an AI-enabled citizen infrastructure reporting platform designed to modernize road maintenance governance across Tamil Nadu.

The proposed solution enables citizens to:

- report potholes,
- damaged roads,
- unsafe infrastructure,
- drainage failures,
- and road safety concerns directly using:
 - mobile devices,
 - photos,
 - GPS tagging,
 - WhatsApp,
 - SMS,
 - or a dedicated mobile application.

The platform creates a transparent, citizen-driven, AI-assisted infrastructure governance ecosystem capable of:

- real-time issue reporting,
- automated departmental routing,

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- SLA-based escalation,
- contractor assignment,
- and executive governance analytics.

The solution supports two citizen engagement models:

1. RoadSnap TN — WhatsApp / SMS Reporting Model
2. RoadSnap TN Mobile Application

The platform is designed for rapid deployment and can launch a pilot within 20 days of approval.

Strategic Outcomes

OBJECTIVE	EXPECTED IMPACT
CITIZEN REPORTING PLATFORM	Faster issue identification
AI-BASED SEVERITY DETECTION	Prioritized repairs
GPS-ENABLED GOVERNANCE	Accurate infrastructure tracking
REAL-TIME SLA MONITORING	Improved accountability
TRANSPARENT ROAD MAINTENANCE	Increased citizen trust
AI-POWERED ANALYTICS	Better infrastructure planning

Delivery Partners

PARTNER	RESPONSIBILITY
VECTRO DIGITAL SERVICES	Platform delivery, UI/UX, deployment, GIS integration, workflow orchestration
EXtream.AI	AI severity classification, geo-analytics, reporting intelligence, operational analytics

2. MANIFESTO ALIGNMENT

The proposal directly aligns with Tamil Nadu's citizen infrastructure modernization vision.

MANIFESTO OBJECTIVE	PLATFORM ALIGNMENT
CITIZEN ROAD REPORTING APP	Photo + GPS reporting system
DIGITAL TAMIL NADU	AI-enabled governance platform
TRANSPARENT GOVERNANCE	Public complaint tracking
SMART INFRASTRUCTURE	Data-driven maintenance planning
REAL-TIME GOVERNANCE	Executive analytics dashboard
AI-ENABLED SERVICES	Severity classification & automation

3. PROBLEM STATEMENT

Current road maintenance systems face several operational and governance challenges.

Citizen Challenges

- No unified reporting mechanism
- Delayed issue resolution
- No complaint visibility
- Manual escalation dependency
- Lack of accountability

Government Challenges

- Fragmented maintenance workflows
- No centralized analytics
- Poor defect prioritization
- Limited SLA tracking
- Reactive maintenance approach

Operational Challenges

- Duplicate complaints
- Contractor coordination delays
- No geo-analytics visibility
- Poor infrastructure intelligence

4. PROPOSED SOLUTION

RoadSnap TN — Smart Citizen Infrastructure Reporting Platform

The proposed ecosystem supports two deployment models.

OPTION 1 — ROADSNAP TN (WhatsApp / SMS Reporting Model)

Concept

“Click. Tag. Fix.”

A lightweight citizen participation model designed for:

- rapid adoption,
- low operational complexity,
- and statewide accessibility.

Citizens do not need to install a dedicated application.

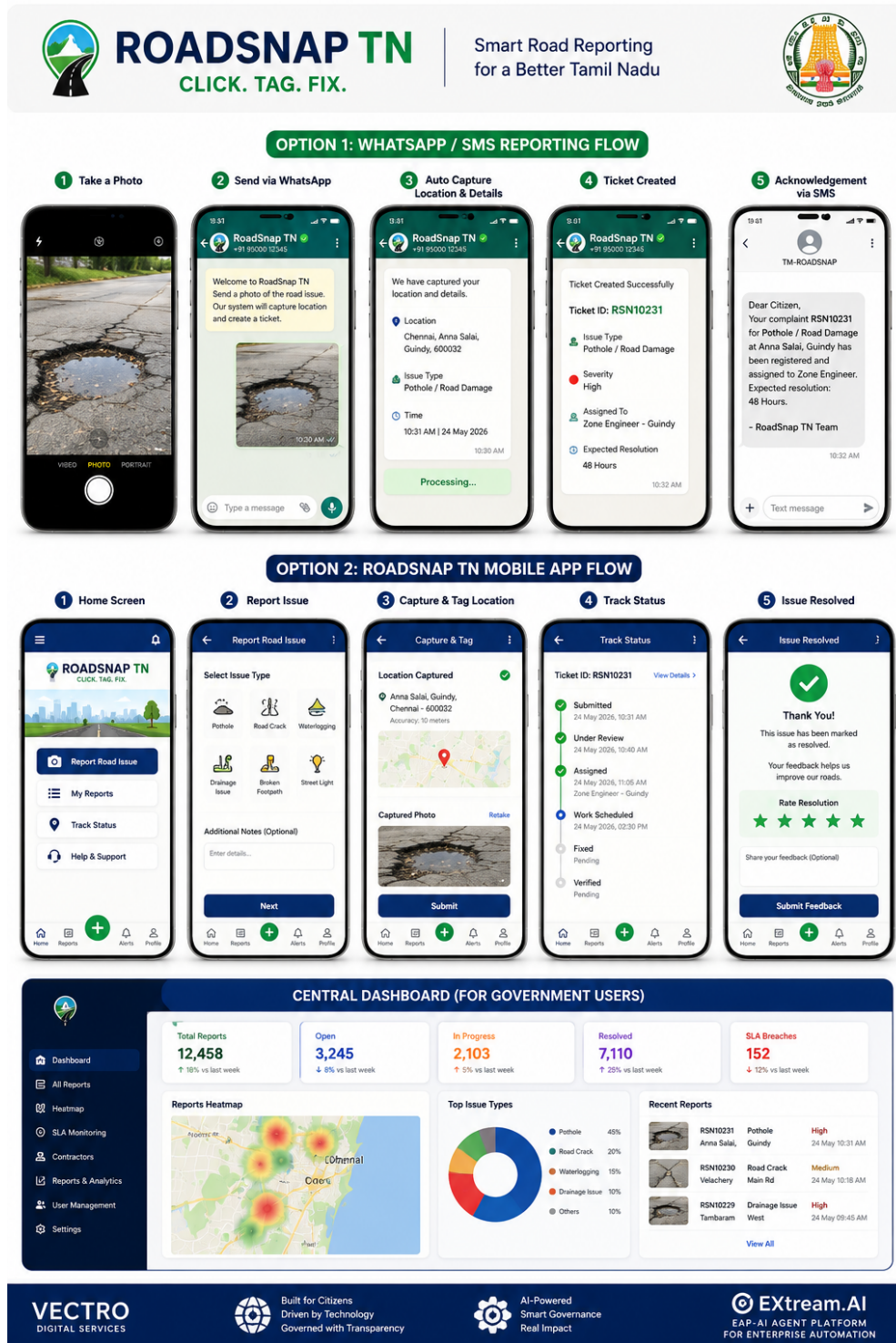
Citizen Workflow

1. Citizen notices road issue
2. Takes photo
3. Sends image via:
 - WhatsApp,
 - SMS,
 - or Telegram channel
4. System captures:
 - image,
 - citizen number,
 - timestamp,
 - geo-location,
 - ward/district mapping
5. AI classifies issue severity
6. Ticket routed automatically
7. Department/contractor assigned
8. Citizen receives SMS acknowledgement

Key Advantages

ADVANTAGE	BENEFIT
NO APP INSTALLATION	High adoption
WORKS ON LOW-END DEVICES	Rural accessibility
WHATSAPP FAMILIARITY	Easy usage
RAPID DEPLOYMENT	Faster rollout
LOW TRAINING REQUIREMENT	Minimal operational overhead

OPTION 2 — ROADSNAP TN MOBILE APPLICATION



AI-Enabled Citizen Governance Platform

The advanced governance version includes:

- dashboards,
- analytics,
- SLA workflows,
- AI engines,
- and contractor governance.

Citizen Workflow

1. Citizen creates profile
2. Selects “Report Road Issue”
3. Takes photo or uploads image
4. GPS location auto-tagged
5. Citizen selects issue category OR AI detects automatically
6. Complaint submitted
7. Citizen tracks lifecycle:
 - Submitted
 - Assigned
 - Work Scheduled
 - Fixed
 - Verified

5. PLATFORM COMPONENTS

COMPONENT	DESCRIPTION
CITIZEN MOBILE APP	Photo uploads, GPS tagging, complaint tracking
WHATSAPP/SMS ENGINE	Lightweight reporting channel
AI CLASSIFICATION ENGINE	Severity detection & duplicate analysis
GOVERNMENT DASHBOARD	Department analytics & complaint management
CONTRACTOR PORTAL	Repair assignment & completion workflows
SLA ENGINE	Escalation & accountability monitoring
ANALYTICS PLATFORM	Heatmaps, trends & infrastructure intelligence

6. CORE FEATURES

6.1 Citizen Features

- Photo uploads
- GPS-based geo-tagging
- Tamil language support

- WhatsApp/SMS reporting
- Complaint tracking
- Resolution notifications
- AI-assisted issue categorization
- Citizen feedback & ratings
- Duplicate complaint detection

6.2 Government Features

- Ward & district dashboards
- SLA monitoring
- Contractor assignment
- Complaint heatmaps
- Real-time governance visibility
- Budget utilization analytics
- Department escalation workflows
- Infrastructure intelligence reporting

6.3 AI Features (EXtream.AI)

AI FUNCTION	PURPOSE
SEVERITY CLASSIFICATION	Prioritize repairs
DUPLICATE DETECTION	Prevent redundant tickets
GEO-ANALYTICS	Identify recurring hotspots
MAINTENANCE FORECASTING	Predict degradation trends
AI SUMMARIZATION	Executive reporting
RESOURCE OPTIMIZATION	Optimize repair scheduling

7. AI & INFRASTRUCTURE INTELLIGENCE

AI Classification Categories

SEVERITY	EXAMPLE
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CRITICAL	Major potholes / unsafe roads
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HIGH	Drainage failures
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MEDIUM	Surface cracks
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LOW	Minor maintenance
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Before & After Validation

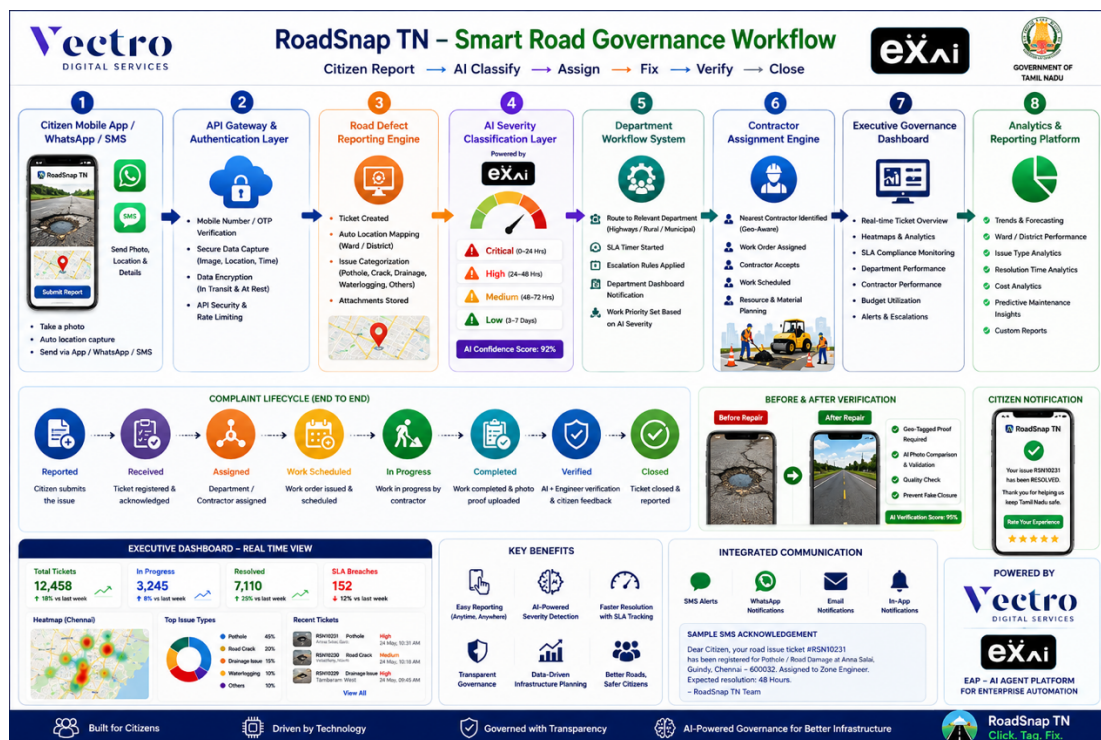
Contractors must upload:

- before repair photo,
- after repair photo,
- geo-tagged completion proof.

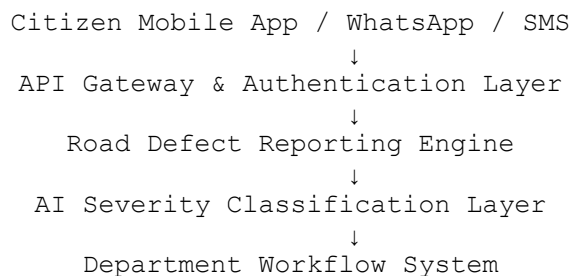
AI validates:

- repair quality,
- location authenticity,
- duplicate closure attempts,
- false completion submissions.

8. TECHNICAL ARCHITECTURE



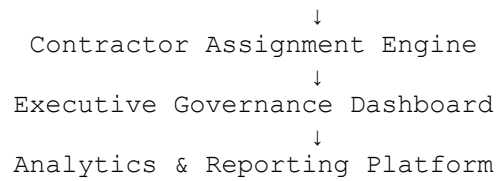
8.1 RoadSnap TN Architecture



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8.2 Technology Stack

LAYER	TECHNOLOGY
MOBILE APPLICATION	React Native
BACKEND APIs	Node.js + Express
DATABASE	PostgreSQL + PostGIS
AI LAYER	EXtream.AI EAPAI Agents
GIS LAYER	OpenStreetMap + GIS APIs
NOTIFICATIONS	Firebase + SMS Gateway
HOSTING	MeghRaj / Azure Government Cloud

9. CITIZEN REPORTING WORKFLOW

Digital Reporting Process

1. Citizen submits road issue
2. GPS location captured
3. AI classifies severity
4. Ticket generated automatically
5. Department routing initiated
6. Contractor assigned
7. Repair completed
8. Citizen notified
9. Feedback collected

10. IMPLEMENTATION ROADMAP

20-Day Pilot Plan

WEEK	ACTIVITIES	DELIVERABLES
WEEK 1	Discovery, branding & GIS integration	BRD + UI mockups
WEEK 2	Mobile app + workflow development	Internal testing build
WEEK 3	AI integration + dashboards	UAT environment
WEEK 4	Security testing & pilot launch	Pilot go-live

Recommended Pilot Zone

Recommended:

- Chennai Corporation Zone
OR
- One district municipality

11. SUPPORT & MAINTENANCE MODEL

TIER	RESPONSIBILITY	SLA
L1 SUPPORT	Citizen support desk	<1 hour
L2 SUPPORT	Platform operations	<2 hours
L3 SUPPORT	AI analytics optimization	<4 hours

Maintenance Activities

- Weekly security patches
- Monthly analytics optimization
- Quarterly platform upgrades
- Annual scalability testing
- 24x7 infrastructure monitoring

Service Availability Targets

METRIC	TARGET
PLATFORM UPTIME	99.5%
CRITICAL RESPONSE	<2 hours
API AVAILABILITY	99.9%
AI PROCESSING ACCURACY	≥90%

12. SECURITY & GOVERNANCE

Security Controls

CONTROL	CAPABILITY
MFA	Secure administrator access
ENCRYPTION	Data protection
AUDIT LOGS	Immutable activity tracking
ROLE-BASED ACCESS	Department-level governance
GPS VALIDATION	Geo-authenticity checks
API SECURITY	OAuth2/API gateway protection

13. RISK MANAGEMENT

RISK	PROBABILITY	IMPACT	MITIGATION
Fake complaints	Medium	Medium	GPS + AI validation
Duplicate reports	High	Medium	AI duplicate detection
Department delays	Medium	High	SLA escalation

High complaint volume	Medium	Medium	AI prioritization
GIS inaccuracies	Low	Medium	Multi-source validation

14. PILOT SUCCESS METRICS

KPI	TARGET
CITIZEN DOWNLOADS	≥ 10,000
COMPLAINTS SUBMITTED	≥ 2,000
SLA COMPLIANCE	≥ 90%
RESOLUTION TIME REDUCTION	≥ 40%
CITIZEN SATISFACTION	≥ 4.2 / 5
DUPLICATE REDUCTION	≥ 60%
PLATFORM UPTIME	≥ 99.5%

15. BOARDROOM STRATEGIC OUTCOMES

The RoadSnap TN platform creates:

- Transparent infrastructure governance
- Citizen-driven maintenance visibility
- AI-enabled infrastructure analytics
- Faster road maintenance cycles
- Executive governance dashboards
- Data-driven budget allocation
- Public trust enhancement

16. ANNEXURES (AVAILABLE ON REQUEST)

- UI/UX Mockups
- GIS Workflow Diagrams
- Technical Architecture Diagrams
- SLA Definitions
- AI Classification Framework
- Security Audit Reports
- Deployment Blueprint

17. NEXT STEPS

1. Steering committee approval
2. Pilot district selection
3. GIS & department workshop
4. Technical kickoff within 5 days
5. Pilot deployment initiation
6. Citizen awareness campaign



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7. Pilot launch within 20 days
 8. Statewide rollout roadmap

CONTACT INFORMATION

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EXtream.AI

Powered by EAP- AI Agent Platform for Enterprise Automation

Together Delivering

Tamil Nadu's AI-Enabled Citizen Infrastructure Governance Platform.

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